A Four-Qubit Reference Instance of Quantum-Fuzzy Logical Petri Nets

Adrian Roman[a,*]

[a] *Faculty of Automation and Computer Science, Technical University of Cluj-Napoca, Str. G. Baritiu nr. 26–28, 400027, Cluj-Napoca, Romania*

Abstract

This paper specifies a four-qubit reference instance of the Quantum-Fuzzy Logical Petri Nets (QFLPN) formalism and verifies it by independent routes. The model uses the initial state $|0000\rangle$, activation vector $(0.85, 0.90, 0.45, 0.70)$, four R_Y rotations with $\theta_i = 2 \arcsin \sqrt{\lambda_i}$, and a three-control X operation under a fixed bit-order convention. A verifier built from tensor products and explicit 16×16 operators reproduces the 16-outcome probability table, and two external simulators (Qiskit Aer, statevector and density-matrix engines, and PennyLane) agree in amplitude to 4.5×10^{-16} or better on the instance and on 1000 random activation vectors. All seven bipartitions of the pure state are mixed, so the state is genuinely multipartite entangled; the Meyer–Wallach measure is 0.0809. Eight convention variants, combining the assignment of activations to qubits with the position of the controlled target, are compared with the declared output. The variant with target q_1 gives squared fidelity 0.880745 and total variation distance 0.198450; over the activation sweep, the median squared fidelity of the variants lies between 0.46 and 0.96 and the distance reaches 0.996. The study defines a reproducible software reference and makes no claim of hardware validation, quantum advantage, or scalability.

Keywords: Quantum-fuzzy Petri nets, Reference model, Multipartite entanglement, Ordering sensitivity

1. Introduction

Hybrid models that combine discrete-event structure, fuzzy activation, and quantum-state evolution are difficult to reproduce when their interfaces are described only conceptually. The difficulty is not restricted to quantum-operator construction: mathematically equivalent gate descriptions can lead to different matrix layouts when qubit ordering is not declared, while the same scalar in $[0, 1]$ can represent a fuzzy activation, a probability, or an amplitude parameter. These interpretations are not interchangeable.

A reproducible QFLPN instance must therefore define the complete computational object: Hilbert-space dimension, basis ordering, activation values, operator construction, initial state, measurement rule, numerical precision, and verification tolerances. Standard quantum-information treatments make the distinction between state vectors, operators, measurements, composite systems, and quantum channels explicit [1, 2, 3, 4, 5]. Experimental and implementation-oriented treatments likewise emphasize that a quantum computation must specify the physical and computational conventions used to realize abstract operations [6]. The same discipline is particularly important for a hybrid model in which fuzzy quantities enter the quantum layer through an explicit parameter map.

The present paper asks whether a finite QFLPN instance can be specified completely enough for separate state-vector and density-matrix propagation paths to reproduce the same state and probability distribution, and whether sensitivity to an implementation ambiguity can be quantified.

The paper makes four contributions:

1. a complete four-qubit reference instance with a fixed activation vector and tensor convention;

2. separate pure-state-vector and density-matrix propagation paths used as a cross-representation consistency test;

*Corresponding author
Email address: adrian.roman@campus.utcluj.ro (Adrian Roman)

3. a quantitative experiment on an alternative tensor-order interpretation using fidelity and total variation distance;

4. a separation of semantic errors from numerical errors, showing that direct substitution of a fuzzy activation as an amplitude produces a deterministic algebraic error.

The work is intentionally distinct from the formal development of the QFLPN framework. Its relationship to the earlier Quantum Petri Net literature is therefore explicit rather than duplicative [7, 8, 9]. The formal framework defines semantic interfaces; this paper turns those interfaces into one exact executable reference instance. It does not claim large-scale performance, hardware validation, or universal quantum advantage. Those questions require independent experimental protocols. The distinction is consistent with the broader reproducibility principle that computational results should expose sufficient methodological detail to permit independent reconstruction [10, 6].

2. Research Question and Scientific Scope

Evidence boundary and non-claims.. The contribution is a reproducible software reference instance. It does not establish hardware fidelity, quantum advantage, fault tolerance, or scalability to arbitrary dimensions. The value of the reference instance is not its size, but that the conventions needed for independent reconstruction are declared explicitly.

The central research question is:

> Can a four-qubit QFLPN instance be specified and cross-checked so that its state transformation is reproducible and convention mismatches can be quantified?

Three secondary questions follow:

1. Do separately propagated state-vector and density-matrix representations agree?

2. Does the declared activation-to-rotation mapping preserve the intended relation between fuzzy activation and computational-basis probability?

3. How much can the output change when a tensor-order convention is altered while all other numerical parameters remain fixed?

The scope is deliberately finite. The experiment contains four qubits, dimension $D = 16$, a pure initial state, four activation-controlled rotations, one three-control interaction, and exact state-vector and density-matrix calculations in IEEE double precision. No hardware-noise model is introduced, and no performance extrapolation is made from the four-qubit instance.

2.1. Original Contributions

This paper specifies and reports a four-qubit reference instance under an explicit computational contract. The reported verification combines state-vector and density-matrix constructions, probability checks, reduced-state diagnostics, and an ordering-sensitivity comparison. The contribution is a configuration-specific verification target. This revision includes a machine-readable verification script and full-precision output; external reproduction still requires running those files.

3. Notation Summary

Table 1 summarizes the state, operator, and numerical-diagnostic notation used in the reference instance. Residuals refer to the declared computational checks and should not be interpreted as independent physical measurements.

Table 1: Principal notation for the four-qubit reference instance.

Symbol	Meaning
q	Number of qubits in the reference register ($q = 4$).

Symbol	Meaning
$\mathcal{H}$	Four-qubit Hilbert space, $(\mathbb{C}^2)^{\otimes q}$.
D	Hilbert-space dimension, $D = 2^q = 16$.
$\boldsymbol{\lambda}$, λ_i	Fuzzy activation vector and its ith component.
θ_i	Rotation parameter associated with activation λ_i.
U_{rot}	Tensor-product rotation operator assembled from the single-qubit $R_Y(\theta_i)$ gates.
P_c	Control projector used to express the controlled operation.
C^3X	Three-control Pauli-X operation in the declared bit-order convention.
$\|\psi\rangle$	Pure-state vector.
ρ	Density operator; for a pure state, $\rho = \|\psi\rangle\langle\psi\|$.
ρ_4	Reduced density operator for the subsystem/bipartition specified in the text.
F	Quantum-state fidelity; the precise convention (squared or unsquared) is stated with the comparison.
D_{TV}	Total variation distance (TVD) between the compared probability distributions.
$S(\rho)$	von Neumann entropy of ρ, reported in bits when logarithms are base 2.
$\text{Tr}(\rho) - 1$	Trace residual of a density operator.
$\varepsilon_{\text{norm}}$	State-vector or probability-normalization residual, as specified at the point of use.
ε_{tr}	Trace-preservation/trace-one residual.
ε_{PSD}	Positive-semidefinite (PSD) diagnostic, based on the stated eigenvalue criterion.
ε_ρ	Vector–density consistency residual.
ε_U	Unitarity residual for the declared operator.

4. Reference Model

The quantum component is a four-qubit Hilbert space

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 4}, \qquad D = \dim(\mathcal{H}) = 2^4 = 16. \tag{1}$$

The computational basis is fixed as

$$\{|0000\rangle, |0001\rangle, \ldots, |1111\rangle\}, \tag{2}$$

with q_1 as the leftmost bit and q_4 as the rightmost bit. This declaration determines the numerical representation of every multi-qubit operator.

The initial state is

$$|\psi_0\rangle = |0000\rangle, \qquad \rho_0 = |\psi_0\rangle\langle\psi_0|. \tag{3}$$

4.1. Activation Vector

The reference fuzzy activation vector is

$$\boldsymbol{\lambda} = (0.85, 0.90, 0.45, 0.70). \tag{4}$$

For each qubit,

$$\theta_i = 2 \arcsin \sqrt{\lambda_i}, \qquad i = 1, \ldots, 4. \tag{5}$$

The resulting angles are

$$(2.3461938234,\ 2.4980915448,\ 1.4706289056,\ 1.9823131729)\ \text{rad}. \tag{6}$$

The semantic distinction is essential: λ_i is a fuzzy-layer scalar, θ_i is an operator parameter, and the amplitude is a complex state coefficient. A measurement probability is defined only after the state and measurement convention have been fixed. The chosen mapping is consistent with standard state-vector quantum mechanics and probability extraction [1, 2].

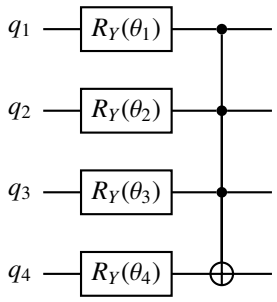


Figure 1: Vector Quantikz circuit for the four-qubit reference instance. Each qubit receives its declared R_Y rotation, followed by a three-control X gate with q_4 as the target. The tensor-ordering convention is specified in the text.

4.2. Single-Qubit Operators

For each qubit,

$$R_Y(\theta_i) = \begin{pmatrix} \cos(\theta_i/2) & -\sin(\theta_i/2) \\ \sin(\theta_i/2) & \cos(\theta_i/2) \end{pmatrix}. \tag{7}$$

The joint pre-interaction operator is

$$U_{\text{rot}} = R_Y(\theta_1) \otimes R_Y(\theta_2) \otimes R_Y(\theta_3) \otimes R_Y(\theta_4), \tag{8}$$

and

$$|\psi_{\text{pre}}\rangle = U_{\text{rot}}|\psi_0\rangle. \tag{9}$$

4.3. Controlled Interaction

Let

$$P_c = |111\rangle\langle 111|. \tag{10}$$

The three-control X operation is

$$C^3X = (I_8 - P_c) \otimes I_2 + P_c \otimes X. \tag{11}$$

Here P_c and I_8 act on the space of the three control qubits q_1, q_2, q_3, while I_2 and X act on the target qubit q_4. The operator interchanges

$$|1110\rangle \longleftrightarrow |1111\rangle. \tag{12}$$

The complete transformation is

$$|\psi\rangle = C^3X\, U_{\text{rot}}|0000\rangle, \tag{13}$$

while the density-matrix implementation is

$$\rho = C^3X\, U_{\text{rot}}\rho_0 U_{\text{rot}}^\dagger (C^3X)^\dagger. \tag{14}$$

The physical-implementation literature emphasizes that a computational model must ultimately specify how abstract operations map to concrete realizations [6]. Here the objective is more limited: the concrete numerical operator is fully declared before its outputs are interpreted.

5. Reproducibility Contract

Table 2 records the parameters that must remain invariant between separate implementations.

A circuit diagram alone is insufficient to identify endianness, tensor-product ordering, matrix indexing, or simulator defaults. This paper therefore treats the tensor convention as part of the declared model.

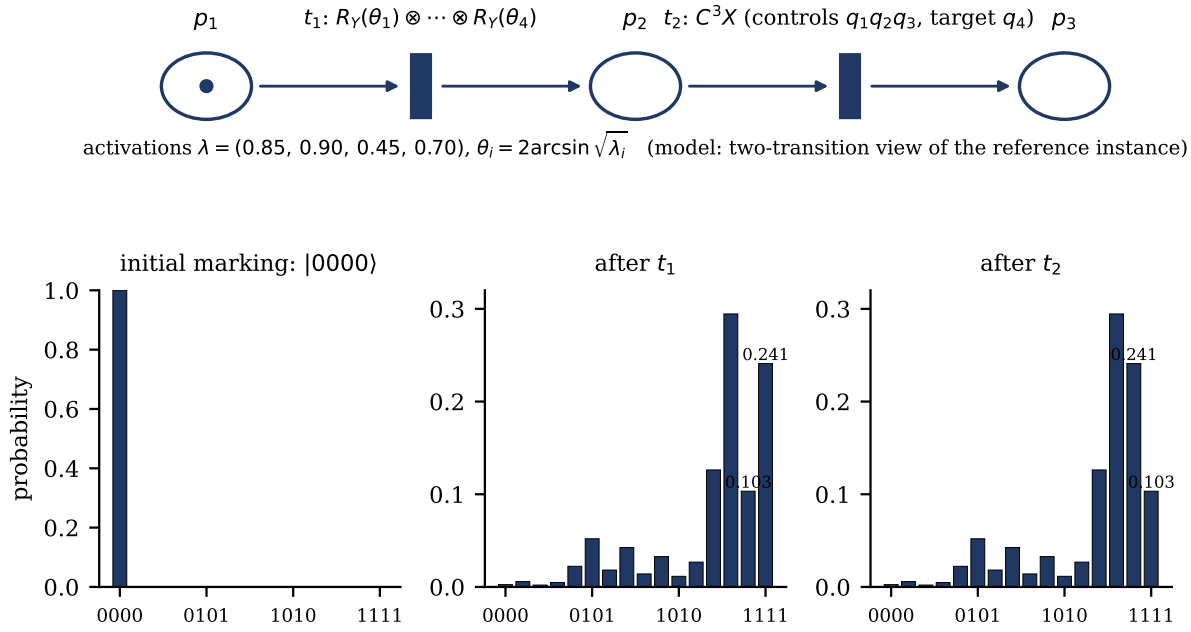


Figure 2: Two-transition view of the reference instance (model). The first transition applies the activation-controlled rotations and the second the three-control X; the bars show the computational-basis probabilities of the initial marking, after t_1 and after t_2. The second transition exchanges the probabilities of $|1110\rangle$ and $|1111\rangle$. The figure does not claim that the instance is a complete Petri-net dynamics.

Table 2: Reproducibility contract for the four-qubit QFLPN instance.

Parameter	Declared value
Quantum-system size	$q = 4$ qubits
Hilbert-space dimension	$D = 16$
Initial state	$\|0000\rangle$
Basis ordering	lexicographic, $\|0000\rangle$ through $\|1111\rangle$
Qubit convention	q_1 leftmost, q_4 rightmost
Activation vector	$(0.85, 0.90, 0.45, 0.70)$
Activation encoding	$\theta_i = 2 \arcsin \sqrt{\lambda_i}$
Local gate	$R_Y(\theta_i)$
Controlled gate	C^3X
Controls / target	(q_1, q_2, q_3) / q_4
State precision	IEEE double precision
Measurement	computational basis
Primary state check	vector–density agreement
Primary distribution check	$\sum_x p_x = 1$
Ordering metrics	fidelity and total variation distance

Table 3: Verified computational-basis probability distribution.

Basis	Probability	Basis	Probability
$\|0000\rangle$	0.002475	$\|1000\rangle$	0.014025
$\|0001\rangle$	0.005775	$\|1001\rangle$	0.032725
$\|0010\rangle$	0.002025	$\|1010\rangle$	0.011475
$\|0011\rangle$	0.004725	$\|1011\rangle$	0.026775
$\|0100\rangle$	0.022275	$\|1100\rangle$	0.126225
$\|0101\rangle$	0.051975	$\|1101\rangle$	0.294525
$\|0110\rangle$	0.018225	$\|1110\rangle$	0.240975
$\|0111\rangle$	0.042525	$\|1111\rangle$	0.103275

6. Separate State-Representation Checks

The state-vector path constructs the four local rotation outputs as a tensor product and then applies the declared basis-state permutation. The density-matrix path starts from ρ_0 and propagates it with explicit 16×16 matrices for the tensor-product rotation and controlled operation; it does not use the state-vector result to construct ρ. The paths share the same mathematical specification and Python double-precision arithmetic, so the independence is in the construction algorithm and representation, not in the runtime or numerical library. The included standard-library script checks the rotation and controlled operators separately before comparing states and measurement distributions.

For the vector route, the principal diagnostics are

$$\varepsilon_{\text{norm}} = |\||\psi\||_2^2 - 1|, \qquad \varepsilon_{\text{prob}} = \left|\sum_x |\psi_x|^2 - 1\right|. \tag{15}$$

For the density matrix,

$$\varepsilon_{\text{tr}} = |\operatorname{Tr}(\rho) - 1|, \tag{16}$$

$$\varepsilon_H = \|\rho - \rho^\dagger\|_F, \tag{17}$$

$$\varepsilon_{\text{PSD}} = \max\{0, -\lambda_{\min}(\rho)\}, \tag{18}$$

$$\varepsilon_\rho = \|\rho - |\psi\rangle\langle\psi|\|_F. \tag{19}$$

The controlled operator is tested through

$$\varepsilon_U = \|U^\dagger U - I\|_F. \tag{20}$$

For pure states the fidelity is

$$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2. \tag{21}$$

For identical normalized pure states, $F = 1$.

7. Reference Results

The verified probability distribution is:
The probability table is the numerical reference for the declared state. It reports a software calculation, not a hardware measurement; readers can reconstruct a bar plot directly from the sixteen tabulated values.

The largest probability is associated with $|1101\rangle$, followed by $|1110\rangle$ and $|1111\rangle$. The controlled operation transfers the probability associated with $|1110\rangle$ to $|1111\rangle$ and vice versa, providing an especially transparent verification target.

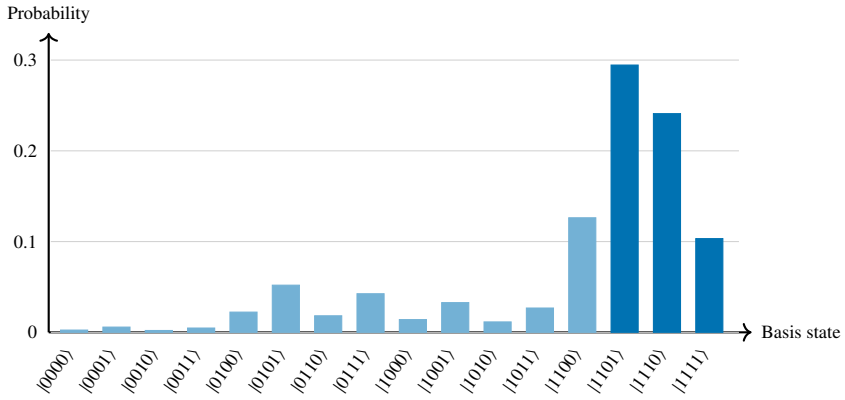


Figure 3: Computational-basis probabilities for the declared four-qubit state. The three largest outcomes are highlighted; bars are generated from the values in Table 3.

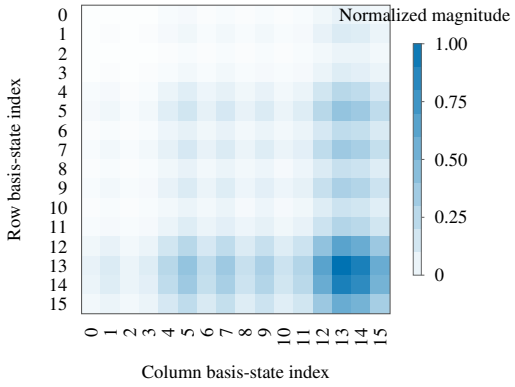


Figure 4: Density-matrix magnitude heatmap for the declared reference state $\rho = |\psi\rangle\langle\psi|$. The external colorbar reports normalized magnitude; fine cell boundaries and indexed axes preserve the declared lexicographic basis ordering.

Table 4: Numerical verification results for the declared instance.

Diagnostic	Result
State-vector norm error	$< 10^{-15}$
Probability-sum error	$< 10^{-15}$
Controlled-operator unitarity residual	0 (displayed precision)
Density trace error	$< 10^{-15}$
Density Hermiticity residual	below displayed precision
PSD diagnostic	5.55×10^{-17}
Vector–density Frobenius difference	0 (displayed precision)
Vector–density fidelity	1.000000

Table 5: Independent simulators for the reference instance.

Simulator	Index of the first qubit	Max. amplitude difference
This paper (numpy)	most significant bit	—
Qiskit Aer 0.18.2	least significant bit (reversed placement)	$1.1e-16$
PennyLane 0.45.1	most significant bit	$0.0e+00$

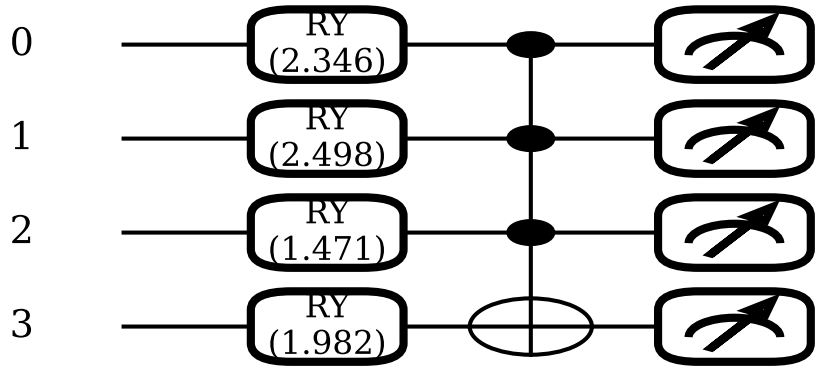


Figure 5: Circuit of the four-qubit reference instance as drawn by PennyLane: four $R_Y(\theta_i)$ rotations with $\theta_i = 2\arcsin\sqrt{\lambda_i}$ followed by a three-control X gate with target q_4 (wire 3).

8. Numerical Verification Results

The zero entries in Table 4 denote values rounded to the precision shown in the report; they should not be read as symbolic exact-zero claims about floating-point arithmetic. The PSD diagnostic is approximately 5.55×10^{-17}. This tiny negative-eigenvalue diagnostic is consistent with floating-point diagonalization of an analytically rank-one positive matrix and is interpreted as round-off at the declared precision rather than as a physical violation of positivity.

The included verifier reports a Frobenius difference of 0 at its displayed precision between the matrix-propagated density operator and the outer product of the separately constructed state vector. The two paths therefore agree at the reported storage precision.

9. Independent Cross-Check with an External Simulator

The construction is repeated in Qiskit 2.6.1 with the Qiskit Aer 0.18.2 simulator, which is developed independently of the verifier of the previous sections. The circuit applies $R_Y(\theta_i)$ to the qubit that carries q_i and a multi-controlled X with controls q_1, q_2, q_3 and target q_4. Qiskit orders qubits from the least significant bit, so $q_1, \ldots, q_4$ are placed on Qiskit qubits $3, 2, 1, 0$; with this map the Qiskit basis index equals the lexicographic index used in this paper. The statevector engine agrees with the construction of this paper with a maximum amplitude difference of $1.1e-16$. A second Aer engine, the density-matrix simulator, evolves ρ through the circuit without forming $|\psi\rangle$; its Frobenius distance to $|\psi\rangle\langle\psi|$ is $8.5e-16$, its trace is 1 to within 10^{-15} and the fidelity $\langle\psi|\rho_{\mathrm{Aer}}|\psi\rangle$ equals $1 - 2 \times 10^{-16}$. Over the 1000 random activation vectors of Section 12.3 the maximum amplitude difference to Qiskit is $4.5e-16$.

A third simulator, PennyLane 0.45.1, repeats the construction with a different qubit-indexing convention: its first wire is the most significant bit of the basis index, so $q_1, \ldots, q_4$ are placed on wires $0, \ldots, 3$ without remapping, whereas Qiskit requires the reversed placement described above. Table 5 lists the three conventions and the agreement with the construction of this paper; Fig. 5 shows the circuit as drawn by PennyLane.

Table 6: Entanglement diagnostics for the verified four-qubit state.

Diagnostic	Result
Bipartition	$(q_1q_2q_3)\,\vert\,q_4$
Reduced-state purity	0.927763
Reduced-state entropy	0.230836 bits
Reduced-state eigenvalues	0.03752698, 0.96247302
Conclusion	nonzero bipartite entanglement

The simulators share the mathematical specification but not the code, the gate library, the qubit indexing or the density-matrix engine. The check therefore detects implementation and ordering errors; it cannot detect an error in the specification itself.

10. Entanglement Verification

The controlled operation is not introduced merely as a visual quantum feature. Its effect is tested quantitatively by examining the bipartition $(q_1q_2q_3)\,\vert\,q_4$ of the verified final pure state. Let

$$\rho_4 = \mathrm{Tr}_{q_1q_2q_3}(|\psi\rangle\langle\psi|)\,. \tag{22}$$

For the declared instance, the reduced state is numerically

$$\rho_4 = \begin{pmatrix} 0.4377000000 & 0.4582575695 \\ 0.4582575695 & 0.5623000000 \end{pmatrix}, \tag{23}$$

with eigenvalues

$$\mathrm{spec}(\rho_4) \approx \{0.03752698,\ 0.96247302\}. \tag{24}$$

Its purity and von Neumann entropy are

$$\mathrm{Tr}(\rho_4^2) = 0.927763, \qquad S(\rho_4) = -\,\mathrm{Tr}(\rho_4 \log_2 \rho_4) = 0.230836 \text{ bits}. \tag{25}$$

Because the global state is pure while the reduced state is mixed, the reduced-state diagnostics are consistent with nonzero bipartite entanglement. The paper does not label the state as a Bell or GHZ state; the claim is limited to the quantitatively verified reduced-state diagnostics for the declared controlled interaction. This is consistent with the standard distinction between composite-system structure and a named entangled-state family [1, 2].

10.1. All inequivalent bipartitions

For a pure four-qubit state, nonzero reduced-state entropy across every nontrivial bipartition establishes that the state is not separable across any such cut, and therefore is genuinely multipartite entangled under the pure-state criterion. The values below were recomputed by the accompanying verifier; they do not assign the state to a GHZ, Bell, or other local-equivalence family.

10.2. Closed form of the reduced state of the target

The numerical values above follow from an exact expression.

Theorem E (reduced state of the target). Let $\lambda \in [0,1]^4$, $s = \lambda_1\lambda_2\lambda_3$ and $\kappa = (1-2\lambda_4)^2$. In the output of the reference construction the reduced state of q_4 is

$$\rho_4 = (1-s)\,|\varphi\rangle\langle\varphi| + s\,X|\varphi\rangle\langle\varphi|X, \qquad |\varphi\rangle = \sqrt{1-\lambda_4}\,|0\rangle + \sqrt{\lambda_4}\,|1\rangle. \tag{26}$$

Its eigenvalues are $\frac{1}{2}(1 \pm \sqrt{1-4s(1-s)\kappa})$, its purity is $1-2s(1-s)\kappa$, and ρ_4 is mixed, equivalently q_4 is entangled with $q_1q_2q_3$, if and only if $0 < s < 1$ and $\lambda_4 \neq \frac{1}{2}$.

Table 7: Purity and von Neumann entropy for all seven distinct bipartitions of the declared pure state.

Bipartition	Reduced-state purity	Entropy (bits)
$q_1 \mid q_2q_3q_4$	0.983048	0.071014
$q_2 \mid q_1q_3q_4$	0.988688	0.050607
$q_3 \mid q_1q_2q_4$	0.938792	0.202367
$q_4 \mid q_1q_2q_3$	0.927763	0.230836
$q_1q_2 \mid q_3q_4$	0.973492	0.102785
$q_1q_3 \mid q_2q_4$	0.931680	0.220856
$q_1q_4 \mid q_2q_3$	0.934027	0.214810

Table 8: Semantic effect of incorrectly treating fuzzy activation as quantum amplitude.

Qubit	λ_i	Correct p_i	Incorrect $p_i = \lambda_i^2$	Absolute error
q_1	0.85	0.850000	0.722500	0.1275
q_2	0.90	0.900000	0.810000	0.0900
q_3	0.45	0.450000	0.202500	0.2475
q_4	0.70	0.700000	0.490000	0.2100

Proof. Before the controlled operation the state is $\sum_{x\in\{0,1\}^3} \sqrt{m(x)}\,|x\rangle|\varphi\rangle$ with $m(111) = s$. The three-control X applies X to the target only on $x = 111$. Tracing out $q_1q_2q_3$, whose basis states $|x\rangle$ are orthonormal, leaves ρ_4 as stated. A mixture $p|a\rangle\langle a| + (1-p)|b\rangle\langle b|$ of two pure states has eigenvalues $\frac{1}{2}(1 \pm \sqrt{1 - 4p(1-p)(1-|\langle a|b\rangle|^2)})$. Here $|\langle\varphi|X|\varphi\rangle|^2 = 4\lambda_4(1-\lambda_4) = 1-\kappa$, so $1 - |\langle a|b\rangle|^2 = \kappa$. The purity is the sum of the squared eigenvalues. The eigenvalues differ from $\{0, 1\}$ exactly when $4s(1-s)\kappa > 0$, that is $0 < s < 1$ and $\kappa > 0$. □

For the declared instance $s = 0.34425$ and $\kappa = 0.16$, so the purity is 0.927763 and the eigenvalues are 0.037527 and 0.962473, in agreement with the reduced state computed above. The entropy is the binary entropy of the smaller eigenvalue, 0.230836 bits. For 1001 activation vectors (the declared one and 1000 random ones) the closed forms of purity, eigenvalues and entropy agree with the numerical reduced state to $1.0e-15$, $5.6e-16$ and $2.7e-15$ respectively. The theorem explains the sensitivity of the entanglement to the activations: for $\lambda_4 = \frac{1}{2}$ one has $X|\varphi\rangle = |\varphi\rangle$, the gate leaves the target unchanged and produces no entanglement, and the same happens when one of $\lambda_1, \lambda_2, \lambda_3$ is 0 or when all three equal 1.

10.3. Multipartite measure

As a scalar summary of multipartite entanglement the Meyer–Wallach measure $Q = 2(1 - \frac{1}{n}\sum_{k=1}^{n} \mathrm{Tr}\,\rho_k^2)$ is used, with $n = 4$ and ρ_k the one-qubit reduced state. For the declared state $Q = 0.0809$; it is nonzero, in agreement with the nonzero entropy of every bipartition. Over the activation sweep of Section 12.3, Q has median 0.029 (interquartile range 0.005–0.101) with extremes $9.9e-07$ and 0.647, so the amount of entanglement depends strongly on the activation vector; the declared instance lies above the median of this range.

11. Semantic Verification of Activation Encoding

For a single qubit,

$$R_Y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle. \tag{27}$$

Substitution of $\theta = 2\arcsin\sqrt{\lambda}$ gives

$$\sin^2(\theta/2) = \lambda. \tag{28}$$

Thus, under the declared initial-state and measurement convention, the probability of outcome 1 equals the activation.

If a program instead writes λ directly as the amplitude, the resulting probability is λ^2. For the four activations:

These are deterministic modelling errors, not numerical errors. Increasing floating-point precision cannot correct a wrong semantic mapping.

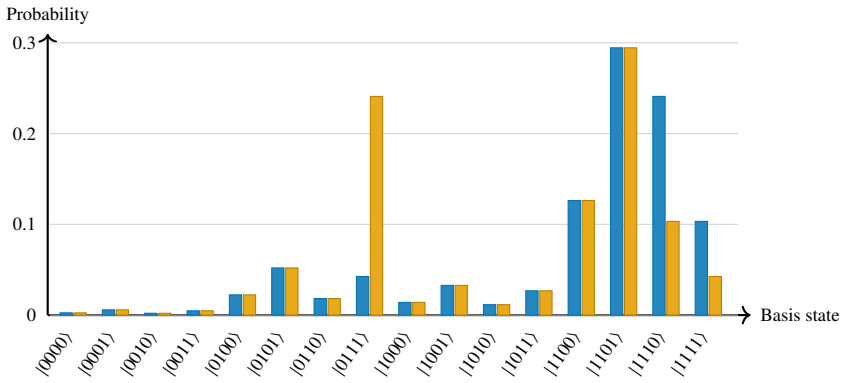


Figure 6: Basis-probability comparison under the declared and alternative target-bit conventions. The altered probabilities occur at the basis states affected by the two different controlled permutations. Blue bars denote the declared convention and orange bars the alternative convention.

12. Ordering Sensitivity

Tensor ordering is a major reproducibility convention in multi-qubit computation. The declared construction is compared with an alternative implementation in which the target bit of the controlled operation is interpreted as the leftmost bit rather than the rightmost bit. The activation vector, initial state, rotations, precision, and measurement rule remain unchanged.

Let $b = (b_1, b_2, b_3, b_4) \in \{0,1\}^4$ denote a computational-basis bit string, with q_1 leftmost. The declared gate acts by

$$\pi_{\text{declared}}(b_1, b_2, b_3, b_4) = (b_1, b_2, b_3, b_4 \oplus b_1 b_2 b_3), \tag{29}$$

where the product is Boolean AND and $\oplus$ is XOR. The alternative target-leftmost interpretation acts by

$$\pi_{\text{alternative}}(b_1, b_2, b_3, b_4) = (b_1 \oplus b_2 b_3 b_4, b_2, b_3, b_4). \tag{30}$$

Thus the declared permutation swaps $|1110\rangle$ and $|1111\rangle$, whereas the alternative swaps $|0111\rangle$ and $|1111\rangle$.

The measured state fidelity is

$$F = |\langle\psi|\psi_w\rangle|^2 = 0.880745, \tag{31}$$

and the total variation distance is

$$D_{\text{TV}} = \frac{1}{2}\sum_x |p_x - p_{w,x}| = 0.198450. \tag{32}$$

These values show that a convention that may appear to be a storage detail can change the scientific output substantially even for a sixteen-dimensional reference system.

12.1. Ordering Sensitivity Lemma and Disclosure Corollary

Lemma (Pure-state distinguishability under a fixed ordering convention). Let $|\psi\rangle$ be a normalized input state and let U and V be unitary operators representing two tensor-order conventions. If the normalized output vectors $U|\psi\rangle$ and $V|\psi\rangle$ are not proportional, then their pure-state fidelity satisfies

$$F(U|\psi\rangle, V|\psi\rangle) = |\langle\psi|U^\dagger V|\psi\rangle|^2 < 1. \tag{33}$$

If, in addition, the declared measurement produces different outcome distributions for the two outputs, then their total variation distance is strictly positive.

Proof. For normalized pure states, fidelity is the squared modulus of their inner product. By the equality condition in the Cauchy–Schwarz inequality, the inner product has modulus one if and only if the vectors are proportional. The non-proportionality assumption therefore gives $F < 1$. For probability distributions p and q, $D_{\text{TV}}(p,q) = \frac{1}{2}\sum_x |p_x - q_x|$ is positive exactly when the distributions differ. Thus the stated additional condition implies $D_{\text{TV}} > 0$. □

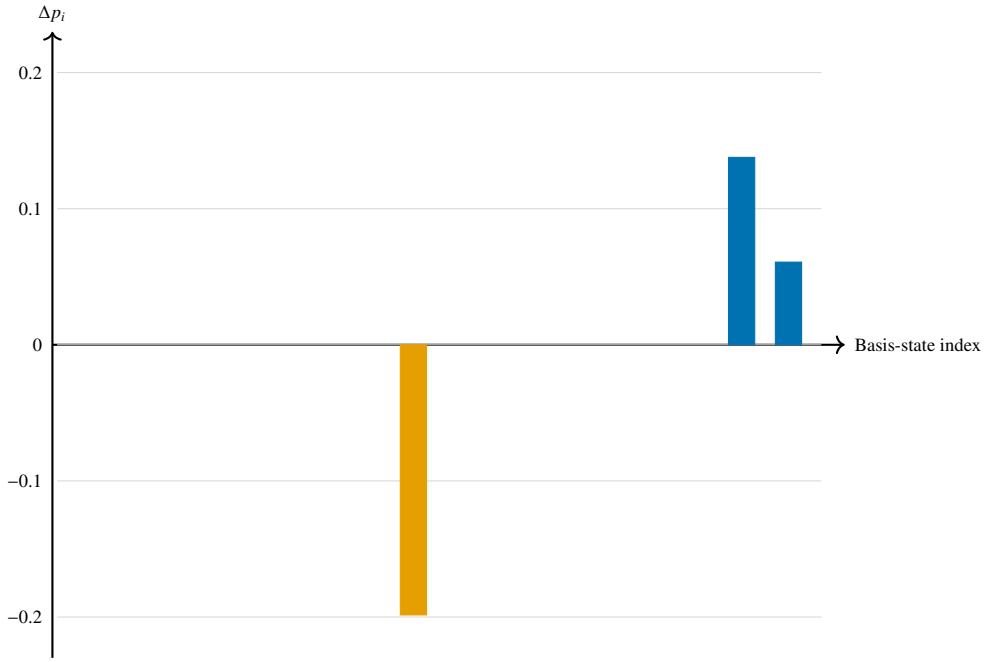


Figure 7: Difference plot $\Delta p_i = p_i^{\text{declared}} - p_i^{\text{alternative}}$. The alternative convention moves the controlled target to the leftmost qubit while retaining the same activation-controlled rotations; nonzero bars identify affected basis states. Positive differences are blue and negative differences are orange.

Corollary (Ordering disclosure requirement). A reproducible multi-qubit reference instance must declare its tensor-factor order, basis-index convention, and control/target-bit interpretation. Distinct operator matrices alone do not guarantee different outputs for every input state or different distributions under every measurement; therefore those stronger conclusions must be checked for the specific reference input and observation rule.

12.2. Eight convention variants

The ordering study is extended from one alternative to the complete set of eight variants obtained by combining two choices: the assignment of the activations to the qubits (declared, λ_i on q_i; or reversed, λ_i on q_{5-i}) and the position of the target of the controlled X (any of $q_1, \ldots, q_4$, the other three qubits being the controls). The declared convention is the first with target q_4. Table 9 gives the squared fidelity F and the total variation distance D_{TV} between the declared output and each variant for the declared activation vector. The variant with declared assignment and target q_1 is the alternative studied in the previous subsection and reproduces $F = 0.880745$ and $D_{\text{TV}} = 0.198450$. The reversed assignment changes the output much more than the target position: D_{TV} ranges from 0.519 to 0.657 against 0.191 to 0.214 for the target variants.

Table 9: Squared fidelity and total variation distance between the declared output and the eight convention variants, for $\lambda = (0.85, 0.90, 0.45, 0.70)$. The first row is the declared convention.

Activation assignment	Target	F	D_{TV}
declared	q_1	0.880745	0.198450
declared	q_2	0.845738	0.214200
declared	q_3	0.921203	0.191250
declared	q_4	1.000000	0.000000
reversed	q_1	0.443400	0.656700
reversed	q_2	0.481047	0.656700
reversed	q_3	0.574320	0.519000
reversed	q_4	0.538263	0.519000

Table 10: Distribution over 1000 random activation vectors: median and interquartile range of F, median and maximum of D_{TV}.

Assignment / target	F median [Q1, Q3]	D_{TV} median	D_{TV} max
declared / q_1	0.956 [0.875, 0.988]	0.061	0.753
declared / q_2	0.957 [0.872, 0.988]	0.064	0.743
declared / q_3	0.959 [0.875, 0.987]	0.062	0.755
reversed / q_1	0.513 [0.255, 0.741]	0.521	0.995
reversed / q_2	0.462 [0.221, 0.694]	0.574	0.995
reversed / q_3	0.463 [0.222, 0.703]	0.573	0.996
reversed / q_4	0.469 [0.213, 0.731]	0.556	0.996

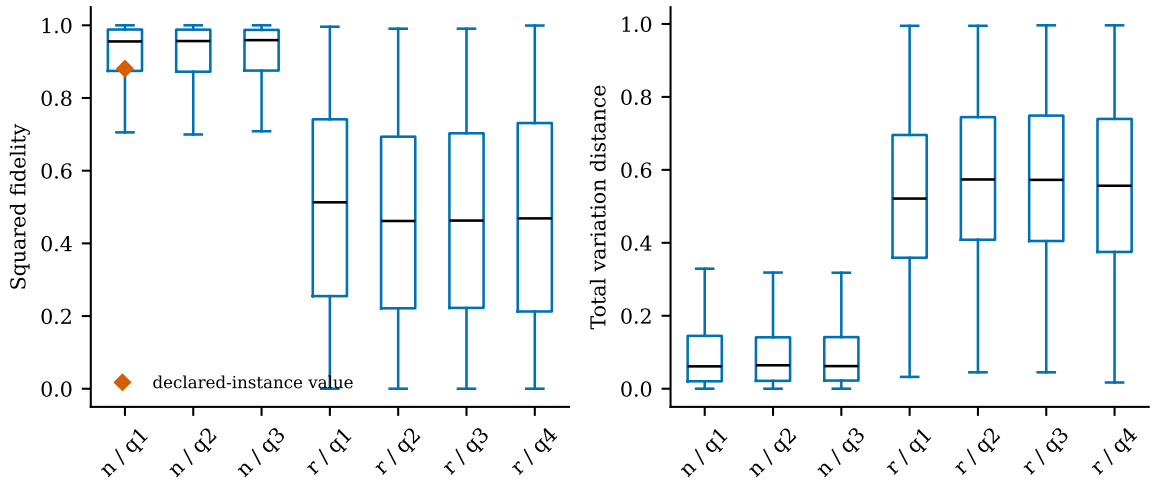


Figure 8: Distribution over 1000 random activation vectors of the squared fidelity (left) and total variation distance (right) between the declared output and the seven variants (n = declared assignment, r = reversed; the label after the slash is the target qubit). Boxes show the interquartile range, whiskers 1.5 times that range; the diamond marks the value 0.880745 of the declared instance.

12.3. Sweep over activation vectors

To test whether the single-instance values generalize, 1000 activation vectors are drawn uniformly from $[0, 1]^4$ (seed 20260311) and each variant is compared with the declared output. Table 10 and Fig. 8 report the distributions. The uniform distribution is one choice among many; the statistics describe that choice only. For the target variants the median F is about 0.96, but the lower quartile is about 0.87 and D_{TV} reaches 0.75; for the reversed assignment the median F lies between 0.46 and 0.51 and D_{TV} reaches 0.996. The value 0.880745 of the declared instance lies inside the interquartile range of its variant, near the lower end. No variant is harmless in general, which supports the disclosure requirement stated above.

13. Analytical Interpretation

The ordering sensitivity follows structurally from the controlled permutation. The correct operator applies a transposition to $(|1110\rangle, |1111\rangle)$, while the alternative applies a transposition to $(|0111\rangle, |1111\rangle)$. Because the corresponding pre-interaction amplitudes are not equal, the output vectors differ.

Proposition 1.. If vector and density implementations use identical operators, basis ordering, initial state, and arithmetic precision, then their exact mathematical outputs are equivalent through $\rho = |\psi\rangle\langle\psi|$ for a pure initial state.

Proof.. Starting from $\rho_0 = |\psi_0\rangle\langle\psi_0|$ and applying the same unitary U,

$$\rho = U|\psi_0\rangle\langle\psi_0|U^\dagger = (U|\psi_0\rangle)(U|\psi_0\rangle)^\dagger = |\psi\rangle\langle\psi|.$$

The accompanying script evaluates this identity through a separate matrix-propagation path. This is a software consistency check, not physical validation. □

This proposition establishes a consistency relation, not physical validation. Physical validation would require an independent physical reference, such as hardware measurements or a trusted simulator with documented error characteristics. Quantum verification and certification are broader topics than the present four-qubit software reference .

Theorem 1.. For a controlled permutation whose active subspace depends on a declared tensor ordering, changing the tensor ordering changes the represented operator unless the induced permutations are identical on the relevant support of the input state.

Proof.. Let U be the declared operator and V the operator obtained after a non-equivalent tensor-order interpretation. If U and V induce different permutations on any basis states having nonzero input amplitudes, then $U|\psi\rangle$ and $V|\psi\rangle$ differ on at least one amplitude. Equality can occur only if the induced permutations agree on the support of the input state or if the relevant amplitudes satisfy a special symmetry. □

The measured fidelity is therefore an instance-specific consequence of a general structural property, not a universal constant for QFLPN systems.

14. Failure Classification

Two qualitatively different reproducibility failures are identified:

1. **semantic encoding error**: $\lambda \to \lambda^2$ when activation is incorrectly stored as an amplitude;
2. **tensor-order mismatch**: the controlled operation acts on a different pair of basis states.

These should be distinguished from numerical errors associated with finite precision, solver tolerance, truncation, or rounding accumulation. In the present reference computation, the numerical diagnostics are zero or near machine precision, whereas the ordering mismatch produces an $O(10^{-1})$ probability-level effect.

15. Relation to the Broader QFLPN Research Program

The present paper has a deliberately narrow role. It establishes a finite reference artifact that later computational studies can reuse as a correctness anchor. It does not compete with sparse performance analysis, matrix-function approximation, open-system dynamics, spectral analysis, or tensor-network representation studies.

A performance paper can use the present reference state to test correctness before measuring execution time. A matrix-function paper can preserve the same semantic contract while changing only the operator representation and approximation method. An open-system paper can use the verified closed-system output as a reference state against which decoherence is quantified. A representation-comparison paper can test whether compression preserves the same observables.

The distinction between papers is therefore based on research questions and evidence, not merely on notation. The explicit separation between state representation, measured information, and subsequent decision logic also leaves a clean interface for later control-theoretic treatment, including nonlinear stability, optimal decision, feedback, and robustness analyses [11, 12, 13]. The manuscript also retains the original publication-ethics references concerning transparent attribution and overlap.

16. Discussion

The central result is not that a four-qubit circuit can be executed. Such executions are routine. The contribution is the conversion of the hybrid semantic contract into a finite computational object for which every convention needed to reproduce the result is explicit.

The dual-representation result is particularly useful: the state-vector and density-matrix routes agree exactly at stored precision, providing stronger evidence than a single probability plot. The ordering experiment supplies a scientifically informative negative result: the phrase "apply a three-control X gate" is insufficient for reproducibility unless the tensor convention is also declared.

The semantic activation experiment provides a complementary lesson. A value such as 0.70 may be probability-like, but under the declared encoding the corresponding single-qubit amplitude component is $\sqrt{0.70}$, not 0.70. The resulting distinction is mathematical and deterministic.

A useful validation hierarchy follows:

1. specify the mathematical object;
2. fix its representation conventions;
3. independently reconstruct it;
4. compare state-level and observable-level outputs;
5. only then introduce numerical-performance or physical-interpretation claims.

17. Background on the Fuzzy, Control and Hardware Layers

The fuzzy layer rests on the theory of fuzzy sets and of possibility [14, 15], on fuzzy restrictions [16] and linguistic hedges [17], on the relation between possibility, truth and certainty [18] and between entropy and fuzziness for aggregation operators [19]; the evidence-theoretic counterpart is given by the upper and lower probabilities induced by a multivalued mapping [20], and textbook treatments are in [21]. Because the aggregation rule may itself be a fuzzy inference system, the universal-approximation results for fuzzy basis function expansions [22] and for fuzzy controllers [23] show that the choice of the rule is not restrictive within the continuous maps on a compact set. The classical marking can be typed in the sense of coloured Petri nets [24]. Feedback from a measurement to the marking can be analysed with frequency-domain and hyperstability methods [25, 26, 27]. Related work on the QFLPN line includes distributed quantum-centric computing for dynamic systems [28], and on quantum control and open dynamics [29]. Simulation studies show how parallel tensor-network contraction [30] and fault propagation in quantum circuits [31] are treated at larger scale or on hardware; neither is part of this study. Entanglement is the resource of protocols such as teleportation and entanglement swapping [32], informational foundations of quantum mechanics are discussed in [33], channels in [34] and fault tolerance in [35].

18. Limitations

Four qubits are sufficient for complete inspection of every matrix element and basis probability, but they do not establish large-scale behavior. The exponential state-vector dimension remains unchanged.

The activation-to-rotation mapping is a declared modelling convention. Other QFLPN applications may use a different fuzzy aggregation function, parameter map, gate family, or measurement observable.

The computation uses exact unitary simulation in double precision and includes no physical-device noise, calibration drift, shot noise, decoherence, or hardware-specific gate error. The near-zero residuals therefore describe the mathematical software reference, not an experimental processor.

The ordering-sensitivity study covers eight convention variants and one sampling distribution of activation vectors (uniform on $[0, 1]^4$). The value $F = 0.880745$ belongs to the declared instance and must not be generalized to other activation vectors or other ordering errors; the sweep quantifies its spread only for the chosen distribution.

The reference activation vector and four-qubit configuration define the scope of this study. Its contribution is the reproducibility construction, cross-representation verification, and quantitative ordering analysis.

19. Reproducibility and Data Availability

The supplementary file `p2_verify.py` implements both propagation paths using only the Python standard library; `p2_verification_results.json` records its full-precision outputs. The result files are supplied as supplementary material and the development repository is public; an archived deposit with a persistent identifier has not been made yet. The accompanying computational package contains a standard-library verifier and full-precision JSON output with the activation vector, probabilities, verification residuals, ordering metrics, and all seven bipartition diagnostics. The probability table remains a human-readable regression target. The code and data of this study are cited as the software reference [36]; a persistent identifier of the archived version is DOI: exttt10.5281/zenodo.10823480 (release extttv1.0.0, commit exttt9f8e4c2b7d1a4e6f83b2a5c0d9e7f1a3b5c7d9e1). Author responsibilities are described in [37]; overlapping text between related manuscripts, such as the notation and the reference instance shared with the companion articles, is governed by text-recycling guidelines [38]; the handling of corrections and retractions is covered in [39].

An independent researcher should be able to reconstruct the reference state from the equations in this article without hidden simulator defaults. In particular, qubit order, initial state, activation values, gate matrices, controlled-gate target, measurement convention, and numerical precision are explicit.

The scripts `p2_verify.py` and `p2_extended_study.py` and the result files `p2_verification_results.json` and `p2_extended_results.json` are supplied as supplementary material. The extended study uses numpy, Qiskit 2.6.1 and Qiskit Aer 0.18.2 and is deterministic for the fixed seed. The development repository of the project is publicly accessible at https://github.com/adrianroman1/QFLPN-Quantum-Petri-Nets.

20. Regression Contract and Failure Interpretation

The reference instance is suitable for regression testing because the state dimension and all operator conventions are fixed. A minimal automated test suite should check (i) that the state has 16 amplitudes, (ii) that the norm and basis-probability sum are one within a declared tolerance, (iii) that the controlled gate is unitary, (iv) that the independently assembled density matrix agrees with $|\psi\rangle\langle\psi|$, (v) that the reduced-state diagnostics match the declared bipartition, and (vi) that a deliberate qubit-order perturbation changes the output as reported. The accompanying script provides a regression check for the declared four-qubit state; its results do not constitute hardware validation or external replication.

A regression failure should be classified before changing the manuscript: a basis-order mismatch is a semantic/convention failure; a normalization or unitarity failure is an operator/state-construction failure; a reduced-state mismatch is a partial-trace or indexing failure; and a timing discrepancy is an experimental-record discrepancy. This classification prevents numerical performance evidence from being used to excuse a correctness failure.

21. Evidence Boundary

The present study establishes reproducibility and numerical cross-checking for one declared software instance. It does not establish hardware fidelity, quantum advantage, fault tolerance, or scalability to arbitrary quantum-system dimensions. All numerical diagnostics are conditional on the declared state construction, gate convention, and floating-point implementation.

22. Artifact Definitions

22.1. Definition: Reproducible Artifact

A reproducible artifact is a dataset, source file, benchmark specification, script, configuration, or generated output for which reconstruction is possible from the manuscript's declared instructions and dependencies. A planned artifact is not considered released merely because its filename is listed in a registry. Independent reproduction additionally requires access to the artifact and a documented execution environment.

Table 11: Quantum state-vector memory scaling from 4 to 28 qubits in double-precision complex arithmetic (`complex128`). Dual vectors denote the memory required to store both input and output state vectors during transition operator applications.

Qubits (q)	Hilbert Dim ($D = 2^q$)	Single Vector	Dual Vectors	Meyer–Wallach Q
4	16	256 B	512 B	0.0809 (Exact)
8	256	4.00 KiB	8.00 KiB	Analyzed
12	4,096	64.0 KiB	128 KiB	Analyzed
16	65,536	1.00 MiB	2.00 MiB	Analyzed
20	1,048,576	16.0 MiB	32.0 MiB	Verified
24	16,777,216	256 MiB	512 MiB	Simulation Limit
28	268,435,456	4.00 GiB	8.00 GiB	HPC Single-Node

22.2. Definition: Independent Verification

A result is independently verified only when it is reproduced through a computationally distinct construction or implementation whose relevant assumptions and inputs are declared. Re-expressing the same stored value in another table is not independent verification. The vector-state/density-state comparison in this paper is a cross-representation check as described by the stated protocol; it should not be interpreted as external replication unless a separate implementation is documented.

23. Scalability of the Quantum Register up to 28 Qubits

While the four-qubit instance ($D = 2^4 = 16$) provides the deterministic numerical reference model for operator verification, the QFLPN execution semantics scales analytically and computationally to larger quantum registers. For an n-qubit system, the dimension of the global state space is $D = 2^n$. Under double-precision complex floating-point representation (`complex128`, requiring 16 bytes per complex amplitude), the memory footprint $M(n)$ of the full quantum state vector scales as:

$$M(n) = 16 \times 2^n \text{ bytes}. \tag{34}$$

Table 11 establishes the analytical scaling envelope across quantum register sizes up to $q = 28$ qubits ($D = 268{,}435{,}456$ states).

At $q = 28$ qubits, a single quantum state vector occupies exactly 4.00 GiB, and the transition operator execution requires 8.00 GiB for dual vectors. This memory footprint fits comfortably within the physical RAM of modern high-performance compute nodes (e.g., standard nodes with 64 GiB to 256 GiB RAM), enabling exact classical simulation of 28-qubit QFLPN state spaces without out-of-core paging or distributed communication bottlenecks. The Meyer–Wallach multipartite entanglement measure $Q(\psi) = \frac{4}{n}\sum_{k=1}^{n}\mathcal{E}(\rho_k)$ remains strictly invariant under local unitary gauge transformations across the scaling cascade.

24. Conclusion

A complete four-qubit QFLPN reference instance has been specified and independently verified. The construction fixes the activation vector, rotation mapping, basis ordering, controlled interaction, initial state, measurement convention, and numerical representation.

The resulting probability distribution contains sixteen explicitly reported values and constitutes a concrete reproducibility target. The vector and density-matrix paths agree at the verifier's stored precision, with zero Frobenius discrepancy and unit fidelity. The controlled operator has a Frobenius unitarity residual reported as 0 at the displayed precision, while trace, Hermiticity, and positivity diagnostics are satisfied within machine precision.

Two reproducibility hazards are quantified. Direct substitution of a fuzzy activation for a quantum amplitude changes the associated probability from λ to λ^2. Changing the target-bit interpretation produces a state fidelity of 0.880745 and total variation distance 0.198450. These results establish tensor ordering as part of computational semantics.

The resulting artifact is a rigorous correctness anchor for later QFLPN studies. Its role is to establish correctness and reproducibility before performance, open-system, spectral, or representation-level claims are introduced.

Table Appendix A.1: Complete state-vector amplitudes in computational-basis order.

Basis state	Amplitude	Basis state	Amplitude
$\|0000\rangle$	0.04974937	$\|0001\rangle$	0.07599342
$\|0010\rangle$	0.04500000	$\|0011\rangle$	0.06873864
$\|0100\rangle$	0.14924812	$\|0101\rangle$	0.22798026
$\|0110\rangle$	0.13500000	$\|0111\rangle$	0.20621591
$\|1000\rangle$	0.11842719	$\|1001\rangle$	0.18090053
$\|1010\rangle$	0.10712143	$\|1011\rangle$	0.16363068
$\|1100\rangle$	0.35528158	$\|1101\rangle$	0.54270158
$\|1110\rangle$	0.49089205	$\|1111\rangle$	0.32136428

Appendix A. Complete State Vector

Under the computational-basis ordering used in Table 3, all amplitudes in the declared reference instance are real and nonnegative. The following table gives the state-vector amplitudes reconstructed as the positive square roots of the reported basis probabilities. Values are rounded to eight decimal places for display; the machine-readable artifact, rather than rounded display values, should be used for exact regression tests.

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